

Statement of Work (SOW)

Project Title: Inventory Management for small Business

Date: 09-02-2026

Prepared By: Team-23

1. Introduction

This Statement of Work outlines the objectives, scope, deliverables, timeline, and responsibilities for the development of an “Inventory Management System for Small Businesses” to be executed by Team-23, as part of the Software Engineering course.

The purpose of this project is to design and develop a software solution that helps small business owners efficiently track inventory levels, manage stock movement, reduce losses, and improve operational decision-making through automation and analytics. The project will be carried out over the **entire semester** and aims to provide hands-on experience in applying software engineering principles to a real-world problem faced by small-scale businesses.

2. Scope of Work

Project Description:

The project focuses on designing and developing a software-based inventory management system that helps small businesses efficiently manage products, stock levels, sales, and supplier details. The system will provide real-time inventory updates, alerts for low stock, and analytical reports to support business decision-making.

Objectives:

The key objectives of the Inventory Management System are:

- Implement Product & Category Management to organize inventory efficiently.
- Enable Stock-In / Stock-Out Tracking for monitoring real-time stock movement.
- Provide Low-Stock Alerts to prevent shortages.

- Offer Supplier Management to maintain supplier details and streamline procurement.
- Maintain Sales & Purchase History for transactional transparency and audits.

Analytics and Decision support:

- Develop a Dashboard with Analytics for quick business insights.
- Include Overstock & Understock Prediction to reduce holding costs and avoid stockouts.
- Provide Sales Trend Visualization for understanding customer demand patterns.
- Support Historical Data Analysis to identify long-term performance trends.

Advanced Operational features:

- Integrate Mobile Product Scanning (Barcode) for faster stock updates and product lookups.
- Enable Branch-Wise Stock Tracking for multi-location management.
- Support Stock Transfer Between Branches with accountability tracking.

User and access control:

- Implement Admin & Staff Roles with controlled permissions to ensure system security and proper workflow management.

Key Activities:

- System architecture and database design
- Module-wise implementation of inventory features
- User interface and user experience design
- Testing, debugging, and validation
- Documentation and final project presentation

3. Deliverables

- **Deliverable 1:** Software Requirements Specification (SRS)
 - **Deliverable 2:** System Design Document (Architecture & Database Design)
 - **Deliverable 3:** Working Inventory Management Application (Core Features)
 - **Deliverable 4:** Final Project Report, User Manual, and Presentation
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4. Timeline and Milestones

Milestone	Description	Due Date
Project Kickoff	Team formation, problem definition, tool selection	Week 1
Phase 1 Completion	Requirement analysis and SRS documentation	Week 2
Phase 2 Completion	System design (UML diagrams, DB schema)	Week 3
Phase 3 Completion	Development of core inventory modules	Weeks 4–6
Testing Phase	System testing, bug fixing, validation	Week 7
Final Delivery	Submission of software, report & presentation	Week 8

5. Roles and Responsibilities

Team Members:

Business Analyst: Likhith Reddy - SE23UARI056

Frontend Developer: Chaithanya Naidu - SE23UARI009, Rihanth – SE23UARI063

Backend Developer: Sohan Varma - SE23UCSE163, Thilak – SE23UCSE099

System Architect: Vishwatej Goud - SE23UARI072

Database Designer: Sindhu Reddy – SE23UCSE091

Testing: Punith – SE23UARI045

Client Contact: Software Engineering course, Mahindra University

6. Budget and Payment Terms

N/A

7. Assumptions and Constraints

Assumptions:

- Project is developed for academic and learning purposes
- Required development tools and software are available
- Continuous guidance will be provided by course instructors

Constraints:

- Academic project deadlines.

- Limited development time alongside other academic commitments
 - Scope restricted to essential inventory management features
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8. Approval Signatures

Client:

Name: _____

Title: _____

Signature: _____

Date: _____

Service Provider:

Name: _____

Title: _____

Signature: _____

Date: _____